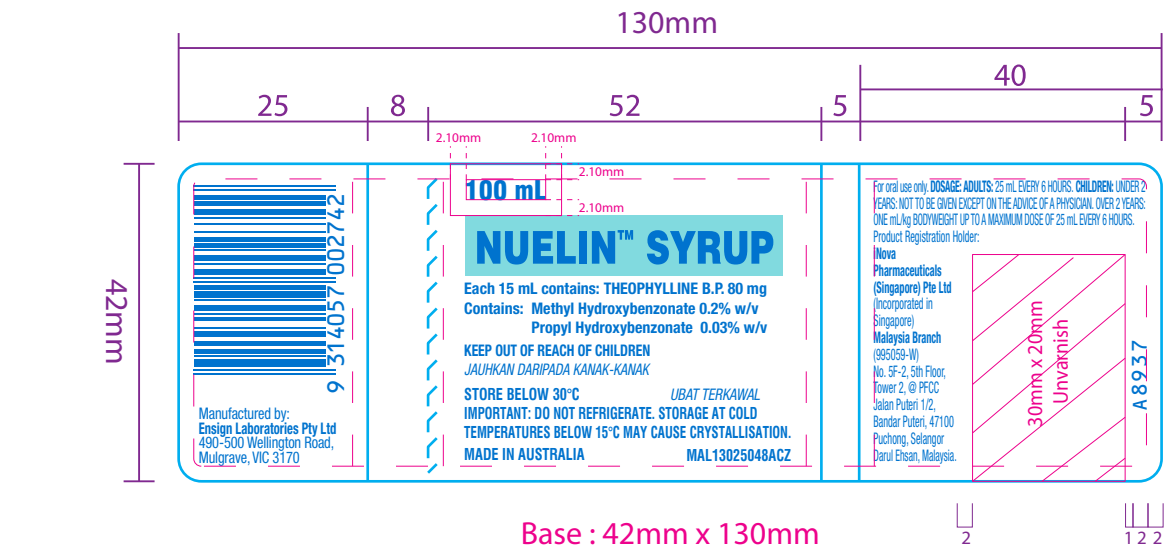
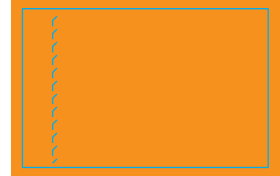
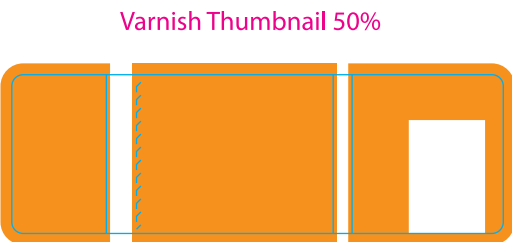
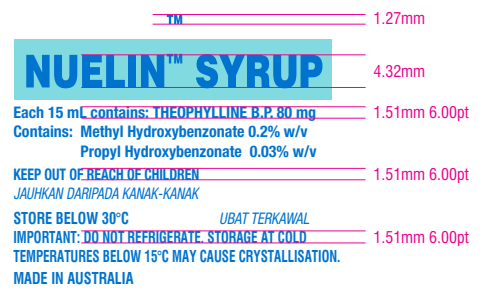
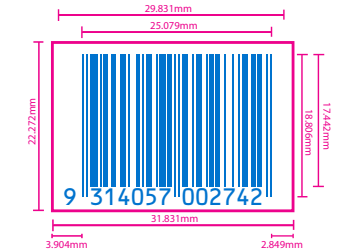




Base : 42mm x 130mm

*All text must be 2mm away from diecut outline and glue area.

80% Barcode Magnification Truncatch



8	57	5	43	47	46	46	47	48	48
100 mL NUELIN™ SYRUP Each 15 mL contains: THEOPHYLLINE B.P. 80 mg Contains: Methyl Hydroxybenzonate 0.2% w/v Propyl Hydroxybenzonate 0.03% w/v KEEP OUT OF REACH OF CHILDREN JAUHKAN DARIPADA KANAK-KANAK STORE BELOW 30°C IMPORTANT: DO NOT REFRIGERATE. STORAGE AT COLD TEMPERATURES BELOW 15°C MAY CAUSE CRYSTALLISATION. MADE IN AUSTRALIA MAL13025048ACZ		The effect of ranitidine, diltiazem, nifedipine, isoniazid, frusemide, influenza vaccine and corticosteroids on theophylline is uncertain, but concomitant use of these drugs should be monitored closely. It is recommended that serum theophylline levels be monitored and dosage adjustments be made if concomitant therapy with these drugs is commenced or ceased during continued theophylline therapy. USE DURING PREGNANCY Theophylline crosses the placental barrier. The effect on foetal development is not known. If theophylline is administered to the mother near the time of delivery, the neonate should be monitored for the		pharmacological effects of theophylline. Premature neonates have a larger volume of distribution and extremely low clearance rates for theophylline. USE DURING LACTATION Theophylline is excreted in breast milk, and is slowly eliminated by infants, particularly if premature. Dosage to the mother should be minimized to avoid toxicity to the infant. ADVERSE REACTIONS The most common adverse reactions are gastric irritation, nausea, vomiting, epigastric pain, tachycardia, palpitations, and tremor. These are usually early signs of toxicity. However, more serious signs of high serum levels (usually above 30 mg/L) such as cardiac arrhythmias and convulsions may appear rarely without prior	warning. As these effects are mostly dose-related the frequency and severity are usually decreased by dose reduction. OVERDOSE Early symptoms of toxicity such as anorexia, nausea, vomiting, headache, irritability and tachycardia may progress to sensory disturbances, confusion, hyperthermia, ventricular arrhythmias, extreme thirst, delirium and convulsion. Treatment There is no specific antidote to theophylline. Symptomatic support is indicated. Gastric lavage should be considered, and if necessary, haemodialysis or peritoneal dialysis.	ROUTE OF ADMINISTRATION Oral DOSEAGE AND ADMINISTRATION Therapeutic plasma levels usually associated with optimal benefit without adverse effects are considered to be between 10 and 20 mg/L (55 to 110 micromol/L). Patient sensitivity may vary however. Variable rates of elimination may necessitate considerable variation in dosage needed to achieve therapeutic levels. Dosage should be individualized, and monitoring of theophylline plasma levels is recommended if side effects appear or if high doses are needed. Adults: 25 mL every six hours. Children: <i>Under 2 years:</i> not to be given except on the advice of a physician. <i>Over 2 years:</i> 1 mL/kg bodyweight (up to a maximum of 25 mL) every six hours.	THEOPHYLLINE MONITORING If side effects appear or if unusually high doses are required, plasma theophylline should be monitored. Blood samples for monitoring should be drawn immediately before administration of the morning dose when the plasma theophylline level is lowest. Another sample should be drawn 4-6 hours after administration of the sustained-release preparation when the theophylline level is at a maximum. PRESENTATION Nuelin Syrup: 80 mg/15 mL (berry flavoured): 100 mL SHELF-LIFE Nuelin Syrup: 2 years STORAGE CONDITIONS Store below 30°C. NUELIN SYRUP - IMPORTANT Do not refrigerate. Storage below 15°C may cause	crystallisation. Manufactured by: Ensign Laboratories Pty Ltd 490-500 Wellington Road, Mulgrave VIC 3170 Revision date: January 2018	LT0051

Leaflet Front : 42mm x 395mm

48	48	47	46	46	47	48	57	8
NUELIN™ SYRUP PRODUCT INFORMATION COMPOSITION Nuelin Syrup: Each 15 mL contains theophylline 80mg. Also contains methyl and propylhydroxybenzoate and sugar 50 g/100 mL. ACTIONS Pharmacology Theophylline has a direct relaxant effect on the smooth muscle of bronchial airways and pulmonary blood vessels, serving as a bronchodilator and pulmonary vasodilator. It also exhibits activities typical of xanthines such as CNS stimulation including the respiratory centre, cardiac stimulation, coronary vasodilation, diuresis, and increased gastric secretion.	The mechanism of action of theophylline in-vivo is not fully elucidated. It is believed to mediate smooth muscle relaxation by inhibition of phosphodiesterase, thus reducing intracellular hydrolysis of cyclic AMP. Theophylline also acts additively or synergistically with B ₂ -adrenoceptor agonists such as catecholamines. These increase intracellular levels of cyclic AMP by stimulation of adenyl cyclase. The clinical significance of this interaction should be considered when theophylline is used concurrently with sympathomimetic agents. There is no evidence that tolerance develops with continued use of theophylline. Pharmacokinetics Absorption: Nuelin Syrup contains theophylline in an alcohol - free vehicle. Peak plasma theophylline levels	occur 1½ to 2 hours after a dose of Nuelin tablets or Nuelin Syrup. Distribution: Approximately 50 - 70% of the circulating theophylline is bound to the plasma proteins. Theophylline partitions into saliva, breast milk and crosses the placental barrier. Metabolism: Theophylline is metabolised in the liver, principally to 1,3-dimethyluric acid with other minor metabolites being 3-methylxanthine and 1-methyluric acid. 3-methylxanthine has some pharmacological activity but less than theophylline. Excretion: Theophylline and its metabolites are excreted by the kidney. About 10% of the administered dose is excreted unchanged. Half-Life: The elimination half-life of theophylline varies considerably. In healthy adults it ranges from 3 to 12 hours. The half-life is shortened by smoking	and is prolonged by cardiorespiratory or hepatic disease and other drugs (see Precautions). In children aged 1 to 9 years the half-life is usually significantly less than in adults. INDICATIONS For the relief and prophylaxis of reversible bronchospasm associated with bronchial asthma, bronchitis, emphysema, and related conditions. CONTRAINDICATIONS Nuelin syrup should not be used where hypersensitivity to its constituents, or to xanthines generally, is known or has been demonstrated. PRECAUTIONS 1. As there is a correlation between plasma levels of theophylline and therapeutic effect, and as patient response can vary considerably due to variable	rates of elimination, monitoring plasma levels in individual patients is recommended (see THEOPHYLLINE MONITORING). Dosage must be individualized if ideal therapeutic effect is to be achieved. 2. Theophylline should be administered with caution to patients with cardiac disease, severe hypertension or hyperthyroidism. 3. Theophylline clearance decreases in patients with congestive heart failure, impaired renal function, acute pulmonary oedema, chronic obstructive pulmonary disease, pneumonia and acute febrile episodes. Clearance is markedly decreased in patients with hepatic cirrhosis. 4. There is some evidence that theophylline exhibits dose-dependent kinetics, at least in sick and elderly patients. Care should be exercised by	titration of dosage requirements in small increments and by monitoring serum theophylline levels. 5. Theophylline clearance is increased in smokers and dosage may need to be adjusted accordingly. 6. Theophylline should not be administered concurrently with other xanthine medications and caution should be exercised when sympathomimetic agents are also part of the regimen. 7. Xanthine containing beverages (eg. tea, coffee, cola, cocoa) may interfere with some serum theophylline assays. DRUG INTERACTIONS The following drugs have been shown to decrease the hepatic clearance of theophylline, thus increasing its serum concentration: Cimetidine, high dose allopurinol, propranolol, macrolide antibiotics (eg.	erythromycin, clarithromycin) quinolone antibiotics (eg. ciprofloxacin and enoxacin), alcohol, oral contraceptives, mexilitene, tacrine, thiabendazole, disulfiram, Interferon alpha and verapamil. The following substances have been shown to increase the hepatic clearance of theophylline, thus lowering its serum concentration: Tobacco or marijuana smoking, phenobarbitone, phenytoin, carbamazepine and rifampicin. Theoretical potential interactions of theophylline with products containing Hypericum perforatum (St John's wort), possibly involving the CYP 1A2 isoform, could result in reduced plasma levels of theophylline. It is recommended that serum theophylline levels be monitored and dosage adjustments be made if concomitant therapy with these drugs is commenced or ceased during continued theophylline therapy.	Ventricular arrhythmias have been reported when halothane is used concurrently with theophylline. Concurrent use of ketamine with theophylline may lower the seizure threshold. Theophylline has been reported to enhance the renal clearance of lithium, thus reducing serum lithium levels. Synergism with adrenaline and other sympathomimetic amines has been reported with theophylline. Concomitant administration of a B ₂ -adrenergic agonist with methylxanthines has resulted in cardiac arrhythmias and sudden death in studies carried out in laboratory animals. The clinical significance of these findings when applied to humans is not known at present.	LT0051

Leaflet Inside : 42mm x 395mm

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RE-TYPESET ARTWORK. Due to the nature of the files received, this artwork has been re-typeset.

	ARTWORK
Job No. : JB-03122 UIC : A8937 SKU : 100500 A8937_NUELIN SYRUP 100ML MY Component : FAF LABEL-LEAFLET Country: MY	13.08.24 - V0 - Artwork build - Koragot 11.03.25 - V1 - Amend - Suphapon 08.04.25 - V1 - Final Artwork - Passanai 20.05.25 - V2 - Amend - Koragot
Dieline Ref: PP-0875 EMO: ENSIGN Printer: Xxxxxx Xxxxxxxxxxxxxx	G247 File Name: 104988_A8937_NUELIN SYRUP 100ML MY_MY_LL_V2
Colour 100%	

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